Ernesto Ramirez

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EDUCATION

University of Notre Dame | Notre Dame, IN — May 2029
B.S. in Electrical Engineering| Concentration in RF & Electromagnetics — *GPA: 3.97/4.00* | *Dean's List*

RESEARCH EXPERIENCE

Wireless Institute / Interference & Coexistence Project | Notre Dame, IN — August 2026 – Present
Undergraduate Researcher | Advisor: Prof. J. Nicholas Laneman

- Leading an independent study of how Wi-Fi interference degrades 2.4 GHz LoRa and FLRC links (Semtech SX1281), with results intended for publication.
- Designed and validated a fully cabled RF test bench with Python control of an ADALM-Pluto SDR injecting varying power and duty cycle 802.11 OFDM bursts while monitoring SIR synchronously.
- Automated sweeps of SIR, duty cycle and data rate, and showed the link beats a collision model's predictions (49% vs 34% PRR at −30 dB SIR).

PROJECTS

137 MHz VHF Satellite Receiver & LRPT Image Decoder — June 2026 – Present

- Designed a 137 MHz VHF receiver front-end PCB integrating a PGA-103+ LNA and a 3-pole, direct-coupled Chebyshev bandpass filter (series-capacitor coupling, 0.10 dB passband ripple) for satellite downlink reception.
- Validated circuit performance and manufacturability through Monte Carlo tolerance analysis in Qucs and supplementary AC simulations in LTspice, accounting for component tolerances and parasitics prior to fabrication.
- Developing a Python-based signal processing pipeline to demodulate QPSK LRPT telemetry from the Meteor-M2 satellite, received via RTL-SDR V4 and a dipole/QFH antenna, to reconstruct downlinked weather imagery from raw IQ samples.

ESP32-Based Wireless RC Car — April 2026 – May 2026

- Designed, built, and programmed a two-wheel-drive RC car and handheld controller from scratch with two-way ESP-NOW telemetry (RSSI, obstacle alerts).
- Wrote asynchronous, non-blocking firmware in C/C++ for both devices to concurrently handle wireless data reception and motor control.

WORK EXPERIENCE

Central Tire & Auto Performance | Albuquerque, NM — August 2023 – May 2025
Service Technician (part-time)

- Serviced ~5 gasoline and diesel vehicles per week, performing maintenance and repairs to manufacturer specifications under strict safety procedures.
- Diagnosed electrical faults using multimeters, OBD-II scan tools, wiring diagrams, and battery/alternator testers, working with senior technicians to trace failures to the faulty component.

LEADERSHIP AND ACTIVITIES

ND Unmanned Aerial Vehicles Club | Notre Dame, IN — August 2026 – Present
Electrical Systems Squad Member

- Developing the electrical system for an autonomous drone with onboard LLM-based decision-making, preparing for the RTX Autonomous Vehicle Competition, Anduril AI Grand Prix, and AI Maritime Maneuver (AIMM) Indiana Collegiate Challenge.
- Working on RF telemetry and communications subsystems, evaluating radio, antenna, and frequency-band options through trade studies.

TECHNICAL AND LANGUAGE SKILLS

Programming: Python, Java, MATLAB, C/C++,
Technical: KiCad, Qucs, LTspice; SolidWorks/Fusion 360; embedded systems; SDRs (Pluto, RTL-SDR, HackRF One).
Language: Fluent spanish.